

JOB HUNTING AND HIRING PROCESS THROUGH A MULTI- CRITERIA JOB MATCHING SYSTEM USING TF-IDF AND LOCATION BASED SCORING

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

This chapter provides an overview of the research study focused on developing a multi-criteria job matching system that enhances the job hunting and hiring process in Antique, Philippines using TF-IDF and location-based scoring. This chapter includes the introduction, background of the study, objectives, conceptual framework, significance, scope and limitations, and definition of terms.

Introduction

In the digital age, the way people search for jobs and how employers look for qualified applicants has been transformed by online platforms, social media, and electronic recruitment systems. Despite these developments, many job hunters and employers in provincial areas like Antique, Philippines still heavily rely on informal channels such as word-of-mouth referrals, physical postings, and generic social media announcements. These traditional and semi-digital practices often result in mismatched opportunities, longer hiring cycles, and underutilization of local talent (Kuhn & Mansour, 2014).

Job matching is fundamentally an information retrieval and decision-making problem. Job seekers typically describe their skills, educational background, and work experience in free-form text, while employers post job descriptions that outline responsibilities, required skills, and qualifications. Without a systematic way of analyzing and comparing these textual descriptions, job portals may only perform simple keyword matching. Such basic methods fail to capture the relevance and importance of specific terms, leading to imprecise recommendations (Manning, Raghavan, & Schütze, 2008). This creates challenges for job

seekers who must manually sift through numerous postings and for employers who struggle to filter and identify the most suitable candidates.

Term Frequency–Inverse Document Frequency (TF-IDF) is a widely used technique in information retrieval and text mining for measuring the importance of words in documents within a collection (Salton & Buckley, 1988). By transforming job descriptions and applicant profiles into numerical vectors and comparing them using similarity measures, TF-IDF can provide a more meaningful assessment of how well a candidate's skills and experiences match a job's requirements. However, job fit is not defined by skills alone. In local contexts such as Antique, geographical proximity and commuting constraints significantly influence the practicality and desirability of job opportunities.

Location-based scoring helps incorporate distance and geographic preferences into the matching process. By evaluating how close an applicant is to a job location, or whether they are within the same municipality or neighboring areas, the system can prioritize opportunities that are more accessible and realistic for both parties (Ye et al., 2011). This is particularly relevant in provinces where transportation options, travel time, and associated costs may limit a job seeker's willingness or ability to pursue certain positions.

In addition to textual relevance and location, other criteria such as salary expectations, work experience, and job type (e.g., full-time, part-time, remote) may influence decision-making. Thus, a multi-criteria job matching system that integrates TF-IDF-based text similarity and location-based scoring, and can be extended to additional decision factors, has the potential to provide more accurate, practical, and context-aware recommendations (Adomavicius & Tuzhilin, 2005).

By focusing on Antique, Philippines, this study aims to develop and evaluate a job matching system that addresses local needs while leveraging established techniques in information retrieval and decision science. The proposed system seeks to assist job hunters in discovering relevant employment opportunities within or near their preferred locations, and to support employers in identifying candidates whose qualifications and proximity best meet their hiring requirements.

Background of the Study

The province of Antique, located in Western Visayas, is primarily characterized by a mix of urbanizing municipalities and rural communities. Economic activities in the province range from agriculture, fisheries, and tourism to services and small-scale enterprises. While employment opportunities exist across various sectors, many job seekers, especially fresh graduates and early-career professionals, face challenges in accessing information about available positions that match their skills and are within reasonable distance from their residence.

Traditional hiring practices in Antique have typically involved walk-in applications, printed announcements, bulletin boards, and interpersonal networks. In recent years, social media platforms such as Facebook have been used informally for job postings and announcements. However, these approaches often lack structure, search capability, and intelligent filtering. Job postings may be scattered across multiple pages or groups, making it difficult for applicants to discover relevant opportunities efficiently. Similarly, employers may receive many unfiltered applications, many of which do not meet the minimum qualifications or are not geographically suitable.

Existing online job portals and general recruitment platforms are usually designed with national or global markets in mind. While they offer keyword search and basic filters (such as job title, category, and region), they may not be optimized for provincial contexts where detailed geographic granularity—such as municipality or barangay level—is crucial. Moreover, simple keyword-based search does not account for the relative importance of specific skills or terms in job descriptions and résumés (Baeza-Yates & Ribeiro-Neto, 2011).

TF-IDF provides a systematic way to address this limitation by quantifying the importance of terms in documents. In the context of job matching, the “documents” can be defined as job descriptions and applicant profiles. Term Frequency (TF) measures how often a word appears in a document, while Inverse Document Frequency (IDF) reduces the weight of common words that appear in many documents. When combined, TF-IDF highlights terms that are both frequent and distinctive, enabling more meaningful similarity computations between jobs and applicants using methods such as cosine similarity (Manning et al., 2008).

However, a purely text-based matching approach overlooks the geographical realities of job placement, especially in regions like Antique where daily commuting between distant municipalities may not be feasible for many residents. Location-based scoring provides a way to integrate spatial information into the matching process. By considering distances between applicant and job locations, or by categorizing proximity (e.g., same barangay, same municipality, within province), the system can assign higher scores to matches that are geographically practical (Liu et al., 2013).

Furthermore, recruitment decisions often involve multiple criteria that go beyond skills and distance. These may include years of experience, education level, job type, and

salary compatibility. A multi-criteria approach allows these different factors to be combined into a single match score using weighted formulas or other decision-making techniques (Saaty, 1980). Such an approach reflects real-world hiring considerations, where decision-makers weigh several aspects simultaneously when evaluating candidates or job options.

Despite the availability of advanced techniques and platforms, there is limited research and documentation on job matching systems that specifically target provincial areas in the Philippines using TF-IDF and location-based scoring. Many systems focus on broader, urbanized labor markets and do not directly address the local conditions, connectivity constraints, and commuting patterns present in provinces like Antique. This gap provides an opportunity to design, implement, and evaluate a localized job matching system that is sensitive to both textual relevance and geographic practicality.

By developing a multi-criteria job matching system using TF-IDF and location-based scoring tailored to Antique, this study aims to provide empirical evidence on the effectiveness of integrating these techniques in improving the relevance and practicality of job recommendations. The findings may serve as a guide for educational institutions, local government units, and organizations in designing digital solutions that better support employment facilitation at the provincial level.

Research Objectives

General Objectives

This study aims to develop a multi-criteria job matching system for job hunting and hiring process in Antique, Philippines by using TF-IDF-based text matching and location-based scoring.

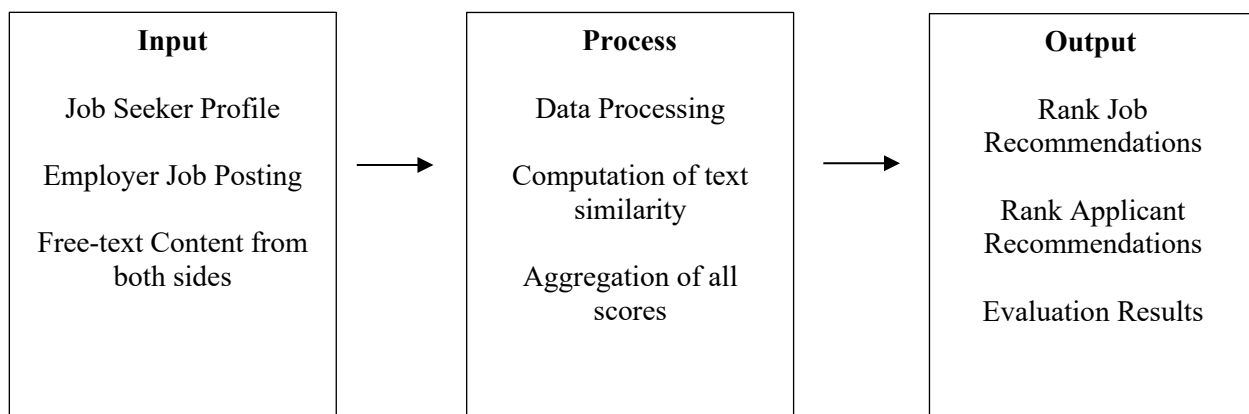
Specific Objectives

Specifically, this study aims to;

1. develop and implement a multi-criteria job matching system that utilizes TF-IDF-based text similarity, location-based scoring, and selected additional criteria to match applicant profiles with job descriptions.
2. implement a ranking and recommendation mechanism that generates prioritized job and applicant matches based on the computed multi-criteria scores, and
3. test the quality of characteristics of the developed system based on ISO 25010 standards.

Conceptual Framework

Figure 1. Input – Process – Output (IPO)



The conceptual framework of this study is based on the Input-Process-Output (IPO) model that illustrates the flow of data and operations in the proposed multi-criteria job matching system using TF-IDF and location-based scoring.

The input stage covers three main components: (1) job seeker data, including personal information, skills, education, work experience, preferred job types, and preferred or current location within Antique; (2) employer job data, including job titles, detailed descriptions and

responsibilities, required skills and qualifications, years of experience, job location, job type (e.g., full-time, part-time, on-site, hybrid), and offered salary range when available; and (3) free-text content from both sides, such as resumes, summaries, skill lists, and detailed job descriptions, which are used for text-based analysis in the matching process.

The process stage consists of three main components: (1) data preprocessing and TF-IDF representation, where applicant profiles and job descriptions are cleaned (e.g., lowercasing, removing punctuation and stopwords, optional stemming or lemmatization) and converted into numerical TF-IDF vectors; (2) computation of text similarity, location-based scores, and additional criteria scores, where the system uses the TF-IDF vectors to measure text similarity between job seekers and job postings, assigns proximity-based location scores, and evaluates other factors such as experience, job type compatibility, and salary alignment; and (3) aggregation of all scores into an overall match score and ranking of job seeker–job pairs, where a weighted formula combines the different scores and produces ranked recommendation lists of the most suitable jobs for each applicant and the most suitable applicants for each job.

The output of the system consists of three main components: (1) ranked job recommendations for job seekers, where job postings are ordered by their overall match score and displayed with key details such as job title, location, and brief justifications (e.g., matched skills, nearby location); (2) ranked applicant recommendations for employers, presenting applicants whose profiles best fit each job posting, along with supporting details such as key matching skills, educational background, and location; and (3) evaluation results, which include user feedback on the relevance of recommendations and comparative analysis between the proposed multi-criteria matching system and a simpler keyword-based baseline.

Significance of the Study

The proposed multi-criteria job matching system is expected to offer benefits to the following:

Job Seekers. For job seekers, particularly fresh graduates and unemployed individuals in Antique, the system can provide more relevant job recommendations that align with their skills, experiences, and preferred locations. By reducing the time and effort needed to manually search through numerous postings, the system may increase their chances of finding suitable employment and improve overall job search efficiency.

Employers and Human Resource Practitioners. Employers, including local businesses, government offices, and organizations, can use the system to identify applicants who closely match their job requirements and are geographically accessible. The ranked applicant lists can help HR practitioners prioritize candidates, streamline screening processes, and potentially reduce the time-to-hire and associated costs.

Educational Institutions. Colleges and universities in Antique, such as the University of Antique, can benefit by leveraging the system as a platform to support graduate placement and industry linkage. By understanding how their graduates are matched to local job opportunities, institutions may also gain insights into skill demand patterns and potential curriculum enhancements.

Local Government Units (LGUs) and Labor Agencies. LGUs and labor or employment offices can utilize the findings of this study to design or improve digital employment facilitation programs tailored to provincial needs. The system's approach to integrating location and skills information can inform policy initiatives aimed at reducing unemployment or underemployment within the province.

Software Developers and System Designers. This study may serve as a reference for software developers and system designers interested in implementing recommendation systems and decision-support tools in the employment domain. The integration of TF-IDF, location-based scoring, and multi-criteria aggregation can provide a practical example of applying information retrieval and decision-making techniques to real-world problems.

Future Researchers. The research will provide a basis for future studies seeking to enhance job matching systems through more advanced natural language processing methods, dynamic weighting schemes, or integration with additional factors such as personality fit or industry-specific competencies. It may also encourage further exploration of employment technologies in other provinces or specialized sectors.

Scope

This study focuses on the design, development, and evaluation of a prototype multi-criteria job matching system that utilizes TF-IDF for text-based similarity and location-based scoring for geographic proximity within Antique, Philippines. The system targets two primary user groups: job seekers and employers operating within the province. Specifically, the scope includes: (1) processing applicant profiles and job postings containing textual information on skills, qualifications, and job descriptions, using TF-IDF and cosine similarity to compute text-based match scores; (2) implementing location-based scoring limited to applicant and job locations within Antique, using municipality or barangay information and, where applicable, approximate distances or categorical proximity levels; (3) developing a multi-criteria model that combines at least text similarity and location-based scores, with possible inclusion of additional criteria such as years of experience or job type compatibility; and (4) evaluating the

system using a limited dataset of sample job postings and applicant profiles collected or simulated for the study, with user feedback and comparative analysis against a basic keyword-based matching approach.

Limitation

The study has several limitations. It does not attempt to cover the entire Philippine job market, as the scope is restricted to Antique province. As a result, findings may not be fully generalizable to highly urbanized or international contexts. The quality of text-based matching depends on the completeness and accuracy of information provided by users; incomplete or poorly written profiles and job descriptions may affect match quality.

The location-based scoring in this study may rely on simplified distance estimations or categorical proximity rather than detailed transportation networks or actual travel times, which could influence the practical accuracy of location recommendations. Additionally, the selection and weighting of criteria in the multi-criteria model will be based on literature, expert opinion, or small-scale surveys, and may not fully capture all preferences of all users.

The prototype system will be developed and tested using available hardware and software resources within the academic setting, which may limit scalability and performance. Integration with existing large-scale job portals, real-time labor market data, or advanced machine learning techniques is beyond the scope of this undergraduate research. Furthermore, the study is conducted within a specific academic year, so future changes in labor market conditions, technologies, or policy environments are not reflected in the system design and evaluation.

Definition of Terms

To provide clarity and facilitate better understanding of key concepts discussed in this study, the following terms are defined:

Applicant Profile. As used in this study, it refers to the collection of information about a job seeker, including personal details, skills, educational background, work experience, preferred job types, and location.

Cosine Similarity. As used in this study, it refers to a measure of similarity between two non-zero vectors that calculates the cosine of the angle between them, used to determine how closely an applicant profile matches a job description based on their TF-IDF representations.

Job Matching. As used in this study, it refers to the process of evaluating and aligning the qualifications, skills, and preferences of job seekers with the requirements, characteristics, and constraints of job openings.

Job Posting. As used in this study, it refers to an employer-created description of an available position, including job title, responsibilities, required skills, qualifications, location, job type, and other relevant details.

Location-Based Scoring. As used in this study, it refers to the process of assigning a numerical score to a job–applicant pair based on their geographic proximity, such as being in the same or nearby municipalities within Antique, Philippines.

Multi-Criteria Job Matching System. As used in this study, it refers to the software system that combines multiple factors—including text-based similarity, location-based scoring, and selected additional criteria—into a single match score to rank job and applicant recommendations.

Relevance. As used in this study, it refers to the degree to which a job opportunity suits an applicant’s qualifications and preferences, or the degree to which an applicant meets the

requirements and context of a job opening, as reflected by the computed match scores.

TF-IDF (Term Frequency–Inverse Document Frequency). As used in this study, it refers to a numerical statistic that reflects how important a word is to a document in a collection, used to convert textual job descriptions and applicant profiles into weighted vectors for similarity computation.

